Chapter 7

Time Series Analysis and Forecasting

7.1 Introduction

Almost every dataset encountered so far in this book, prices in Chapter 2, financial statement line items in Chapter 4, credit spreads in Chapter 5, has a time dimension: observations arrive in a sequence that usually carries information a model ignoring order would throw away.

This chapter makes time itself part of the model. It introduces the statistical vocabulary for describing dependence across time, stationarity, autocorrelation, and volatility clustering, and the classical models built on that vocabulary: autoregressive and moving-average models, ARIMA, GARCH, and state-space models.

Two worked examples run through the chapter: a short, hand-computable interest-rate-deviation series used to fit and forecast an autoregressive model, and the AssetA return series first introduced in Chapter 2, reused here to illustrate how GARCH captures volatility clustering. The chapter also introduces formal stationarity testing, seasonal models, and two ways of extending the analysis to several related series, cointegration and vector autoregression, and it keeps a promise made in Chapter 6: explaining why cross-validation must be modified for time series data.

The chapter follows the order in which an analyst works through a new series: establish whether it is stationary (Section 7.2.3) and how strongly it depends on its past (Section 7.2.4), then fit and forecast a model, with the forecast's uncertainty rather than a single number (Section 7.3.1). It then widens the lens to several series, cointegration (Section 7.4.1) for pairs that move together and the Kalman filter (Section 7.5.2) for a state that is only noisily observed, and closes with the walk-forward validation on which Chapters 8 and 13 depend.

7.2 Foundations of Time Series

Financial series violate the independence every earlier chapter assumed, and this part develops the vocabulary for saying how.

7.2.1 Why Time Series Analysis Matters in Finance

Financial data are overwhelmingly time series: daily prices, quarterly earnings, monthly inflation, tick-by-tick trades. Three features make them harder to work with than the typical machine learning dataset. First, observations are not independent: today's return is related to yesterday's, if only weakly, violating the independent-and-identically-distributed assumption behind many statistical procedures. Second, the variance of returns changes over time, with quiet periods following quiet ones and volatile periods following volatile ones, a phenomenon called volatility clustering. Third, the underlying process generating the data can change over time (Chapter 6's non-stationarity concern), so a model fit on one period may not describe the next.

Time series methods exist to handle these features rather than assume them away, and they underpin much of the rest of this book: the risk models in Chapter 8 need a volatility forecast, the portfolio methods

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in Chapter 12 need a covariance forecast, and the trading strategies in Chapter 13 need to know whether a signal is transient noise or a persistent pattern.

Both this chapter's models and the regression and classification models of Chapter 6 fit parameters by minimizing a loss, but a standard machine learning model assumes each training observation is drawn independently from the same distribution, the assumption a time-ordered dataset violates. A model that ignores the dependence, for example by shuffling a time series before splitting it into training and test sets, produces badly misleading performance estimates. Every technique in this chapter, from the correlogram to GARCH to walk-forward validation, exists to model, exploit, or correctly evaluate a model in the presence of this dependence.

7.2.2 Stationarity, White Noise, and Random Walks

Suppose an analyst fits a seemingly solid model relating an interest rate to its own past values, only to see it fall apart when new data arrive, not because the estimation was wrong but because the series never behaved consistently enough to model. This section makes precise the consistency a series needs for a fitted model to generalize. A time series $\{X_t\}$ is weakly stationary if its mean and variance do not change over time and the covariance between X_t and X_{t-k} depends only on the lag k, not on t itself:

$$\mathbb{E}[X_t] = \mu, \qquad \mathrm{Var}(X_t) = \sigma^2, \qquad \mathrm{Cov}(X_t, X_{t-k}) = \gamma_k \text{ for all } t. \tag{7.1}$$

Stationarity matters because most time series models, including every model in this chapter, assume that relationships estimated from the past will continue to hold; without it, there is no reason to expect a fitted model to generalize.

White noise is the simplest stationary process: a sequence ε_t with mean zero, constant variance σ^2, and zero covariance at every nonzero lag, $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for $k \neq 0$. White noise is unpredictable by construction and serves as the building block, the "noise," in every model that follows. A stronger notion, strict stationarity, requires the entire joint distribution of any collection of observations, not just their mean and variance, to be unaffected by shifting them in time; the models in this chapter rely on weak stationarity, which is easier to verify and sufficient for the estimation methods used.

A random walk is $P_t = P_{t-1} + \varepsilon_t$, and it is not stationary: its variance grows with t (specifically $\mathrm{Var}(P_t) = t\sigma^2$ if P_0 is fixed), so its statistical behavior changes over time. Asset prices behave approximately like random walks, which is why Chapter 2 worked with returns, $R_t = (P_t - P_{t-1})/P_{t-1}$, rather than price levels: returns are much closer to stationary, since they do not inherit the growing variance of the price level. This matters: fitting a stationary-series model such as AR or ARMA to a non-stationary price level produces meaningless, unstable estimates, so the first step in almost any time series analysis is to check whether the series needs to be differenced (transformed to $\Delta X_t = X_t - X_{t-1}$) or converted to returns before modeling.

7.2.3 Testing for Stationarity: The Augmented Dickey-Fuller Test

Deciding whether a series is stationary should not rest on visual inspection alone; the standard formal test is the (augmented) Dickey-Fuller test. The test rewrites the AR(1) model

$$X_t = c + \phi X_{t-1} + \varepsilon_t$$

in first-difference form,

$$\Delta X_t = \alpha + \gamma X_{t-1} + \varepsilon_t, \qquad \gamma = \phi - 1, \tag{7.2}$$

and tests the null hypothesis $H_0 : \gamma = 0$ (equivalent to $\phi = 1$, a unit root, meaning the series is a non-stationary random walk) against the alternative $H_1 : \gamma < 0$ (meaning $\phi < 1$, so the series is stationary and mean-reverting). The "augmented" version adds lagged difference terms

$$\Delta X_{t-1}, \Delta X_{t-2}, \ldots$$

to the right-hand side to account for additional autocorrelation, but the core logic, testing whether γ is significantly negative, is unchanged.

This section and the two that follow use a single running example, Table 7.1: eight monthly observations of a short-term interest rate's deviation from its policy target, in basis points. The series is short enough that

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Month t	1	2	3	4	5	6	7	8
Deviation X_t	9.0	8.0	7.0	7.0	5.0	6.0	4.0	4.0

Table 7.1: A short interest-rate deviation series (basis points)

every calculation can be checked by hand; Section 7.2.4 computes its correlogram and Section 7.3.1 fits an AR(1) model to it.

Applying the (non-augmented) test to this series means regressing ΔX_t on X_{t-1}:

$$\hat{\gamma} = -0.290, \qquad SE(\hat{\gamma}) = 0.259, \qquad t\text{-statistic} = \frac{\hat{\gamma}}{SE(\hat{\gamma})} \approx -1.12.$$

Notice that $\hat{\gamma} = -0.290$ is consistent with the AR(1) coefficient fitted in Section 7.3.1, since

$$\hat{\phi} - 1 = 0.710 - 1 = -0.290$$

exactly. The Dickey-Fuller test does not use standard normal or t-distribution critical values, because under the null hypothesis of a unit root the test statistic has a nonstandard distribution; the relevant critical value at the 5% level (with a constant term and no trend) is approximately -2.89, not the familiar -1.96.

Since $|-1.12| < 2.89$, the test fails to reject a unit root, even though the AR(1) coefficient of 0.71 and the decaying correlogram in Figure 7.1 suggest mean reversion. The contradiction is a consequence of sample size, not a flaw in the test: with only 8 observations, the Dickey-Fuller test has little power to distinguish a stationary process from a unit root, the "limited effective sample size" challenge flagged later in this chapter. On the much longer series to which it is normally applied, its power is far greater.

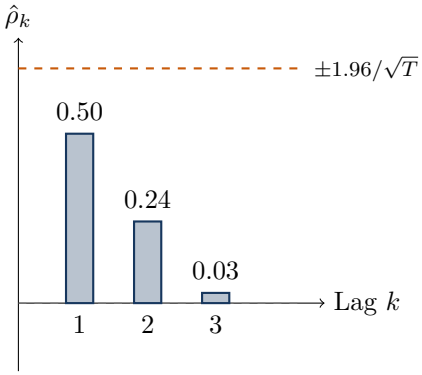


Figure 7.1: Sample correlogram for the interest-rate deviation series in Table 7.1, with the approximate 95% significance band $\pm 1.96/\sqrt{T}$ (dashed; symmetric about zero, so only its positive side is drawn)

7.2.4 Autocorrelation and the Correlogram

Suppose an analyst believes an interest-rate series is mean-reverting rather than a random walk and wants to know how far back its memory extends: does today's value mainly reflect yesterday's, or does a shock from several periods ago still leave a trace? The autocorrelation function (ACF) answers this by measuring how strongly a series is related to its own past values. At lag k, it is defined as

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\mathrm{Cov}(X_t, X_{t-k})}{\mathrm{Var}(X_t)}, \tag{7.3}$$

and is estimated from a sample of T observations with mean $\bar{X}$ as

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^{T}(X_t - \bar{X})^2}.$$

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A plot of $\hat{\rho}_k$ against k is called a correlogram, one of the most common diagnostic tools in time series analysis: white noise has a correlogram indistinguishable from zero at every lag, while an autocorrelated series shows a pattern, decaying gradually for an autoregressive process and cutting off sharply for a pure moving-average process. Figure 7.1 shows the correlogram of the running example.

Because $\hat{\rho}_k$ is estimated from a finite sample, a common rule of thumb treats it as indistinguishable from zero if it falls within an approximate 95% band of $\pm 1.96/\sqrt{T}$, where T is the number of observations. For the 8-observation series this band is $\pm 1.96/\sqrt{8} \approx \pm 0.693$, larger than the sample autocorrelation $\hat{\rho}_1 = 0.497$ computed in the next section; Figure 7.1 plots the band with the correlogram.

Even the largest autocorrelation in the running example is therefore not statistically distinguishable from zero, purely because the sample is so small. This does not contradict the AR(1) pattern the correlogram shows; like the ADF test's failure to reject a unit root, it is a reminder that with 8 observations no time series diagnostic can be fully trusted, and that the qualitative pattern (geometric decay) is more informative here than any single significance test.

7.3 Forecasting Models

This part builds the classical forecasting families in order, from a single autoregressive term to the full ARIMA specification.

7.3.1 Worked Example 7.1: Fitting and Forecasting an AR(1) Model

This section works through the lifecycle of a small time series model: fitting it, checking that its correlogram behaves as theory predicts, and producing a point forecast whose uncertainty the following sections quantify. An autoregressive model of order 1, AR(1), expresses the current value as a linear function of the immediately preceding value plus noise:

$$X_t = c + \phi X_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \text{white noise}(0, \sigma^2). \tag{7.4}$$

The process is stationary if and only if $|\phi| < 1$, in which case its theoretical autocorrelation decays geometrically, $\rho_k = \phi^k$, and its long-run (unconditional) mean is $c/(1-\phi)$.

The data are the running series introduced in Section 7.2.3, the eight monthly interest-rate deviations of Table 7.1.

The sample autocorrelations, computed directly from the formula above using the full-sample mean $\bar{X} = 6.25$, are $\hat{\rho}_1 = 0.50$, $\hat{\rho}_2 = 0.24$, and $\hat{\rho}_3 = 0.03$, plotted in Figure 7.1. The roughly geometric decay, each lag's autocorrelation a bit less than half the previous one, is the signature of an AR(1) process.

The AR(1) coefficients are estimated like any linear regression, by regressing X_t on X_{t-1} with the OLS formulas of Chapter 6:

$$\hat{\phi} = \frac{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})(X_t - \bar{X}_{\text{cur}})}{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})^2}, \qquad \hat{c} = \bar{X}_{\text{cur}} - \hat{\phi}\,\bar{X}_{\text{lag}},$$

where $\bar{X}_{\text{lag}}$ and $\bar{X}_{\text{cur}}$ are the means of the lagged and current values, respectively (each computed over the 7 available pairs, so they differ slightly from the full-sample mean $\bar{X} = 6.25$ used for the correlogram). Fitting this to Table 7.1 gives

$$\hat{\phi} = 0.71, \qquad \hat{c} = 1.19, \qquad R^2 = 0.60.$$

Since $|\hat{\phi}| = 0.71 < 1$, the fitted process is stationary, with an implied long-run mean of $\hat{c}/(1-\hat{\phi}) = 1.1935/0.2903 \approx 4.11$ basis points. This differs noticeably from the sample mean of 6.25: with 8 observations the AR(1) estimate carries substantial uncertainty, and the implied long-run mean is no more precise than the data behind it. The one-step-ahead forecast, given the final observation $X_8 = 4.0$, is

$$\hat{X}_9 = \hat{c} + \hat{\phi}\, X_8 = 1.19 + 0.71(4.0) \approx 4.03 \text{ basis points.}$$

Every autoregressive forecast works this way: the fitted relationship, applied to the most recent observation, produces the next point forecast.

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7.3.2 Confidence Intervals and p-Values for Model Parameters

A point estimate such as $\hat{\phi} = 0.71$ is half the story; without knowing how precisely $\hat{\phi}$ is estimated, one cannot say whether the mean reversion is a feature of the process or an artifact of a short, noisy sample.

Section 6.3.2 built the standard-error, t-statistic, p-value, and confidence-interval machinery, including how (and how not) to state what a confidence interval means. It applies unchanged, since $\hat{\phi}$ is an OLS slope of X_t on X_{t-1}. For the AR(1) regression fitted above, with $n = 7$ usable pairs and 2 estimated parameters (so $n - 2 = 5$ residual degrees of freedom),

$$SE(\hat{\phi}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})^2}} \approx 0.259, \qquad \hat{\sigma}^2 = \frac{\text{SSR}}{n-2} \approx 1.187, \qquad t = \frac{\hat{\phi}}{SE(\hat{\phi})} = \frac{0.71}{0.259} \approx 2.74,$$

where SSR is the sum of squared residuals from the fitted line. Under the null hypothesis $H_0 : \phi = 0$ (this month's deviation carries no information about next month's), this t-statistic follows a t-distribution with 5 degrees of freedom and gives $p \approx 0.041$: by the conventional 5% threshold, $\hat{\phi} = 0.71$ is barely significant, modest evidence of persistence. The corresponding 95% confidence interval,

$$\hat{\phi} \pm t^*_{0.975,\,5} \cdot SE(\hat{\phi}) = 0.71 \pm (2.571)(0.259) \approx (0.04,\ 1.38),$$

is wide, spanning nearly the whole range from very weak to very strong mean reversion and including values close to 1 (the unit-root boundary).

This is the small-sample fragility behind the Dickey-Fuller result of Section 7.2.3, which failed to reject a unit root despite $\hat{\phi} = 0.71$: a t-statistic, a p-value, and a confidence interval are three views of the same estimation uncertainty, which with 8 observations is large. Applying the identical method to the intercept gives $SE(\hat{c}) \approx 1.750$, $t \approx 0.68$, and $p \approx 0.526$, so the intercept is not statistically distinguishable from zero at conventional significance levels, a much weaker result than the slope's.

Section 6.3.2's two warnings apply unchanged: statistical significance is not economic significance, and testing many coefficients or models against the same data raises the chance that one crosses the 5% threshold by luck. The second is the multiple-testing problem that Section 7.4.1's discussion of pairs selection takes up: a p-value below 0.05 is far less persuasive when it is the best of twenty candidate pairs than when it is the only pair considered.

7.3.3 Multi-Step Forecasting and the Growth of Uncertainty

The one-step-ahead forecast above answers "what happens next period," but many financial decisions require forecasting several periods ahead. For an AR(1) model, this is done recursively: use the one-step forecast $\hat{X}_9$ as if it were the actual next observation to produce $\hat{X}_{10} = \hat{c} + \hat{\phi}\hat{X}_9$, and continue in the same way for as many steps as needed. Applying this to the fitted model ($\hat{\phi} = 0.71$, $\hat{c} = 1.19$) starting from $X_8 = 4.0$ gives the forecasts in Table 7.2.

Horizon h	1	2	3	5	10
Forecast $\hat{X}_{8+h}$	4.03	4.06	4.07	4.09	4.11

Table 7.2: Multi-step-ahead AR(1) forecasts from month 8

Two features of Table 7.2 illustrate general properties of stationary autoregressive forecasts. First, the forecasts converge toward the long-run mean of $\hat{c}/(1 - \hat{\phi}) \approx 4.11$ as the horizon grows; a stationary model's best guess about the distant future is its unconditional mean, since the information in today's deviation fades geometrically at rate $\hat{\phi}$ per period.

Second, the uncertainty around these forecasts also grows with the horizon but, unlike a random walk, does not grow without bound: the variance of the h-step forecast is proportional to $\sum_{j=0}^{h-1} \hat{\phi}^{2j}$, which converges to $1/(1 - \hat{\phi}^2) \approx 2.01$ times the one-step forecast variance as $h \to \infty$, rather than growing linearly with h as it would for a non-stationary series. This quantifies the warning in this chapter's challenges section that longer-horizon forecasts are less reliable, showing how much less reliable they are and why, for a mean-reverting series, the unreliability levels off.

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7.3.4 Diagnosing the Fit: The Ljung-Box Test and Choosing Between AR(1) and AR(2)

After fitting, the residuals should be checked for structure the model failed to capture, the diagnostic step of the Box-Jenkins methodology introduced later in this chapter. The Ljung-Box test formalizes this check: it tests the joint null hypothesis that the first h residual autocorrelations are all zero, meaning an adequately specified model should leave behind nothing but white noise, using the statistic

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{n-k}, \tag{7.5}$$

where $\hat{\rho}_k$ is the residuals' own sample autocorrelation at lag k (computed just as in Section 7.2.4, but applied to the residuals rather than the raw series) and n is the number of residuals. Under the null of a correctly specified model, Q follows a chi-squared distribution with $h-p-q$ degrees of freedom, where p and q are the numbers of AR and MA parameters estimated, subtracted because fitting them already absorbs some of the dependence.

Applying this to the AR(1) model fitted above ($\hat{\phi} = 0.71$, $\hat{c} = 1.19$), the seven residuals are

$$(0.42, 0.13, 0.84, -1.16, 1.26, -1.45, -0.03),$$

giving residual autocorrelations $\hat{\rho}_1 = -0.68$ and $\hat{\rho}_2 = 0.49$, both considerably larger in magnitude than a well-fitting model's residuals should show. With $h = 2$ lags and $p = 1$ AR parameter, $Q = 7.91$ against a chi-squared distribution with $2 - 1 = 1$ degree of freedom, giving $p \approx 0.005$, comfortably below the conventional 5% threshold. The AR(1) residuals are not white noise: some structure remains unmodeled.

Does a higher-order model capture more of it? Fitting an AR(2) model,

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t,$$

to the same six usable observations ($t = 3, \ldots, 8$; the sample must shrink by one further observation to accommodate the second lag, and the AR(1) model is refit on this identical six-observation sample so the two models' fit statistics are directly comparable) gives $\hat{\phi}_1 = 0.03$, $\hat{\phi}_2 = 0.88$, and a residual sum of squares of only 1.39, versus 5.60 for the AR(1) model refit on the same six observations, a dramatic improvement. Comparing the two models with the information criteria introduced in the ARIMA section below,

$$\mathrm{AIC} = n\ln(\mathrm{SSR}/n) + 2k, \qquad \mathrm{BIC} = n\ln(\mathrm{SSR}/n) + k\ln n, \tag{7.6}$$

with $n = 6$ throughout ($k = 2$ and $k = 3$ parameters for AR(1) and AR(2), respectively), gives

$$\begin{aligned} \mathrm{AIC}_{\mathrm{AR(1)}} &= 3.59, & \mathrm{AIC}_{\mathrm{AR(2)}} &= -2.76, \\ \mathrm{BIC}_{\mathrm{AR(1)}} &= 3.17, & \mathrm{BIC}_{\mathrm{AR(2)}} &= -3.39 : \end{aligned}$$

both criteria favor AR(2) by a wide margin, even after the extra parameter's complexity penalty. The conclusion deserves caution. An AR(2) model on six observations uses half the sample as estimated parameters ($k = 3$ against $n = 6$), the "limited effective sample size" and "model selection versus overfitting" concerns of this chapter's challenges section, so the much lower AIC/BIC is at least as likely to reflect a small sample fitting its own noise as a better-specified model. On a longer series the workflow, fit a candidate model, test whether its residuals are white noise, and if not compare a higher-order alternative by an information criterion, remains standard practice; only the tiny sample makes this particular conclusion (prefer AR(2)) less trustworthy than the procedure.

7.3.5 Moving Averages and Exponential Smoothing

Two much simpler forecasting tools remain widely used because of their transparency. A simple moving average forecast uses the average of the most recent m observations as the forecast for the next period,

$$\hat{X}_{t+1} = \frac{1}{m}\sum_{i=0}^{m-1} X_{t-i}.$$

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It is easy to compute and understand, but it weights all m past observations equally and drops old observations abruptly once they fall outside the window.

Exponential smoothing addresses both issues by weighting all past observations, but with exponentially decaying weight on older data:

$$\hat{X}_{t+1} = \alpha X_t + (1-\alpha)\hat{X}_t, \qquad 0 < \alpha < 1,$$

which can be expanded to show that $\hat{X}_{t+1}$ is a weighted average of every past observation, with weight $\alpha(1-\alpha)^i$ on the observation i periods back. A larger α reacts quickly to new information at the cost of more noise; a smaller α is smoother but slower. Applied to Table 7.1 with $\alpha = 0.4$ and an initial forecast $\hat{X}_1 = X_1 = 9.0$, the recursion tracks the series' decline without reacting sharply to any single observation, falling from 9.0 to a forecast for the next month of $\hat{X}_9 = 0.4(4.0) + 0.6(5.396) \approx 4.84$. The AR(1) model, by contrast, ties the forecast to the estimated mean-reversion coefficient $\hat{\phi}$ rather than only to recent history.

Because simple exponential smoothing forecasts with only a smoothed level, it always predicts a flat line, which performs poorly for a series with a clear trend, the kind of gradual decline visible in Table 7.1. Holt's linear trend method addresses this by smoothing two components separately, a level L_t and a trend T_t:

$$L_t = \alpha X_t + (1-\alpha)(L_{t-1} + T_{t-1}),$$
$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1},$$

and forecasts h steps ahead as $\hat{X}_{t+h} = L_t + hT_t$, explicitly projecting the estimated trend forward rather than assuming it disappears. Applying this to Table 7.1 with $\alpha = 0.4$, $\beta = 0.3$, an initial level $L_1 = 9.0$, and an initial trend $T_1 = X_2 - X_1 = -1.0$, the recursion, updated once for each of X_2 through X_8, produces a final level of $L_8 \approx 3.631$ and trend of $T_8 \approx -0.706$, giving a one-step-ahead forecast of

$$\hat{X}_9 = L_8 + T_8 \approx 3.631 - 0.706 \approx 2.93.$$

This is well below both the simple exponential smoothing forecast (about 4.84) and the AR(1) forecast (4.03, Section 7.3.1), because Holt's method extrapolates the downward trend instead of assuming the series reverts toward a stable mean, as AR(1) does, or stays flat, as simple smoothing does. Which is more appropriate depends on whether the series is trending or is a mean-reverting process passing through a low patch, a judgment no formula resolves and the question the stationarity testing of Section 7.2.3 addresses: a trending (non-stationary) series suits Holt's method, a mean-reverting (stationary) one the AR(1) approach. A further extension, Holt-Winters smoothing, adds a third seasonal component to handle the kind of periodic patterns discussed in the SARIMA section below.

7.3.6 ARIMA Models

Suppose a shock, say a one-time data-reporting glitch, influences a series for exactly one more period and then vanishes, rather than decaying gradually forever as the AR(1) model implies. The AR(1) model of Section 7.3.1 belongs to a broader family that captures this kind of finite-memory dependence, among others. A moving-average model of order q, MA(q), expresses the current value as a function of recent noise terms rather than recent levels,

$$X_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q},$$

and combining autoregressive and moving-average terms gives an ARMA(p, q) model. Because many financial series (prices, but also many macroeconomic series) are non-stationary in levels but stationary after differencing, the ARIMA(p, d, q) model applies an ARMA(p, q) model to the series after differencing d times: $d = 1$ removes a linear trend, $d = 2$ removes a quadratic trend, and so on. In practice, d is rarely more than 1 or 2 for financial data, since returns (a $d = 1$ difference of log prices) are already close to stationary.

The MA(q) autocorrelation structure is the mirror image of the AR(p) one, and this contrast is what makes the ACF and PACF useful diagnostics together. For an MA(1) process $X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$, the theoretical autocorrelation is $\rho_1 = \theta/(1+\theta^2)$ and $\rho_k = 0$ for every lag $k > 1$; for $\theta = 0.6$, this gives $\rho_1 = 0.6/1.36 \approx 0.441$ and then an abrupt drop to zero at lag 2, in sharp contrast to the AR(1) correlogram in Figure 7.1, whose autocorrelations decay gradually across every lag rather than cutting off sharply. This is the basis of the Box-Jenkins identification rule: a correlogram (ACF) that cuts off sharply after lag q suggests an MA(q)

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model, while a partial autocorrelation function that cuts off sharply after lag p suggests an AR(p) model; a series where both decay gradually suggests a mixed ARMA(p, q) model, for which the specific orders are typically chosen by comparing information criteria across a small number of candidates rather than by reading the correlogram alone.

Choosing the orders p, d, and q is traditionally done with the Box-Jenkins methodology: examine the ACF and its partial-autocorrelation counterpart (the PACF, which measures the correlation between X_t and X_{t-k} after removing the influence of the intervening lags) to identify candidate orders, fit several candidate models, and compare them using an information criterion such as AIC or BIC, which penalize complexity so that the model with the most parameters is not chosen by default. This is the time series version of the model-selection problem of Chapter 6: more lags always fit the training data at least as well, but too many lags overfit sample noise as a high-degree polynomial does.

Many financial and economic series also exhibit seasonality, a repeating pattern at a fixed period, such as retail sales spiking every December or trading volume dipping every summer. SARIMA (Seasonal ARIMA) extends ARIMA by adding a second set of autoregressive, differencing, and moving-average terms operating at the seasonal period s, denoted SARIMA$(p, d, q)(P, D, Q)_s$: the non-seasonal part (p, d, q) captures short-run dynamics as before, and the seasonal part $(P, D, Q)_s$ captures dependence at lags $s, 2s, 3s, \ldots$. A monthly retail sales series might be modeled as SARIMA$(1, 1, 1)(1, 1, 1)_{12}$, the seasonal terms relating each month to the same month one year ($s = 12$ months) earlier on top of the month-to-month dynamics.

Seasonal differencing, $\Delta_s X_t = X_t - X_{t-s}$, is the seasonal analogue of the ordinary differencing introduced above, and it is applied when a series exhibits a seasonal unit root, an extension of the stationarity testing in Section 7.2.3 to the seasonal frequency. Most asset return series show little seasonality at the daily frequency, so this chapter's GARCH and ARIMA examples need no seasonal extension, but seasonality is common in macroeconomic series (retail sales, employment, seasonally-adjusted GDP) that feed into the fundamental analysis of Chapter 4 and the macroeconomic risk factors of Chapters 8 and 9.

7.3.7 Worked Example 7.2: The Same Diagnostics on a Real Treasury Series

Every test applied to the eight observations of Table 7.1 was inconclusive, which says more about the sample than about the tests. On a long real series the same tools do their job. The series is the month-end 10-year U.S. Treasury par yield from January 1990 to December 2023, $T = 408$ months, taken from the same Treasury file that supplies the yield-curve panel of Section 7.4.3; over the period it ranges from 0.55% to 9.04%. Table 7.3 collects the results for the yield and for its monthly change.

Statistic	Yield level	Monthly change
Dickey-Fuller t, no lags (p-value)	-2.158 (0.222)	-18.120 (< 0.0001)
Augmented Dickey-Fuller t, AIC lags (p-value)	-2.488 (0.118)	-11.281 (< 0.0001)
Autocorrelations, lags 1, 2, 3	0.986, 0.970, 0.955	0.099, -0.090, 0.033
Partial autocorrelations, lags 1, 2, 3	0.988, -0.088, 0.053	0.099, -0.101, 0.055

Table 7.3: Stationarity tests and correlograms for the month-end 10-year Treasury yield, January 1990 to December 2023 ($T = 408$; the 5% Dickey-Fuller critical value is -2.87 and the 95% autocorrelation band is ± 0.097)

Stationarity. Regressing ΔX_t on X_{t-1} exactly as in Section 7.2.3 gives a Dickey-Fuller statistic of -2.158 for the yield level, short of the 5% critical value of -2.87, and adding the three lagged differences that AIC selects moves it only to -2.488 ($p = 0.118$). The unit-root null is not rejected, just as it was not for the level factor in Section 7.4.3. This time the failure carries information, because the same test on the monthly changes rejects emphatically, at -18.12 without lags and -11.28 with the two lags AIC selects, far beyond any conventional critical value. With 400 observations rather than 8 the test has the power to tell the two cases apart, and its verdict is that the level behaves like a random walk while its changes are stationary: the diagnose-then-difference step of the workflow at the end of this chapter, carried out on real data.

Correlogram and identification. The 95% band is now $\pm 1.96/\sqrt{408} = \pm 0.097$, seven times narrower than the ± 0.693 of Section 7.2.4. The level's autocorrelations decay so slowly that they are still 0.910 at lag 6, the signature of a series at or near a unit root. Its partial autocorrelations, by contrast, cut off after the first

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lag: 0.988 at lag 1, then −0.088, 0.053, and −0.092 at lags 2 to 4, all inside the band. By the Box-Jenkins rule of the ARIMA section this identifies an AR(1). The monthly changes look much closer to white noise: their first autocorrelation, 0.099, and second partial autocorrelation, −0.101, sit just outside the band and everything else inside it, a little short-run structure that is real but too small to forecast with profitably.

The AR(1) fit. Fitting the AR(1) to the level by OLS, as in Section 7.3.1, gives $\hat{c} = 0.048$, $\hat{\phi} = 0.986$, and $SE(\hat{\phi}) = 0.0065$, with $R^2 = 0.983$. The standard error is forty times smaller than the 0.259 of the eight-point fit, and the naive 95% interval, $0.986 \pm 1.96 \times 0.0065 = (0.973,\ 0.999)$, appears to exclude a unit root. It does not, and this is the trap the Dickey-Fuller test exists to avoid: $(\hat{\phi} - 1)/SE(\hat{\phi}) = -2.158$ is the Dickey-Fuller statistic itself, and under a unit root it must be compared with −2.87, not with −1.96. Read as a stationary model, the fit implies a half-life of $\ln 0.5 / \ln 0.986 \approx 49$ months for a shock to the yield, another way of saying that 34 years of monthly data cannot distinguish very slow mean reversion from none. The high R^2 is equally uninformative, since it mostly records that this month's yield is close to last month's. The eight-point series failed every test for lack of data; the Treasury series passes or fails them for reasons the data can support.

7.4 Multivariate Methods

One series is rarely the whole story; this part models several jointly, and extracts the few factors that drive them.

7.4.1 Cointegration and Pairs Trading

Individual price series are typically non-stationary. A remarkable exception arises when two non-stationary series move together closely enough that a linear combination of them is stationary, a relationship called cointegration. Two stocks in the same industry might each wander like a random walk, yet the spread between them revert reliably to a stable relationship, because both are driven by the same industry fundamentals.

Table 7.4 shows six days of prices for two hypothetical bank stocks.

Day	1	2	3	4	5	6
Stock A	50	51	53	52	54	55
Stock B	48	49	50	50	51	52
Spread $(A - B)$	2	2	3	2	3	3

Table 7.4: Prices for two hypothetical cointegrated bank stocks

Both price series trend upward and would likely fail a stationarity test on their own. The spread, however, fluctuates narrowly around a mean of $\bar{s} = 2.5$ with a sample standard deviation of only 0.55, far more stable than either price. The standard formal test, the Engle-Granger two-step method, first regresses one price on the other to estimate the cointegrating relationship,

$$A_t = \alpha + \beta B_t + u_t,$$

which for this data gives $\hat{\alpha} = -12.5$ and $\hat{\beta} = 1.3$, and then applies the Dickey-Fuller test from Section 7.2.3 to the regression residuals $\hat{u}_t$: if the residuals reject the unit-root null (that is, they are stationary), the two series are cointegrated. One detail matters here: the test statistic must be compared with Engle-Granger critical values (tabulated by MacKinnon), not with the −2.89 used for a single observed series. Because $\hat{\beta}$ was chosen by least squares to make the residuals as small as possible, they look more stationary than an arbitrary series would, so the bar is higher: about −3.34 at the 5% level for two series in large samples, and more negative still in short ones.

The regression generalizes the simple $A_t - B_t$ spread (which assumes $\beta = 1$) to an estimated hedge ratio $\hat{\beta}$, the number of shares of B needed to offset one share of A; the simple difference is a reasonable approximation only when the two series move roughly one-for-one. Here $\hat{\beta} = 1.3$ is close enough to 1 for the raw spread in Table 7.4 to look stable, but, as the z-score below shows, not close enough to trade on.

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This relationship is the foundation of pairs trading, a strategy covered in more depth in Chapter 13. In practice the spread is standardized into a z-score, $z_t = (s_t - \bar{s})/\hat{\sigma}_s$, so that trading rules can be stated in standard deviations rather than raw units, which vary from pair to pair. Having estimated the hedge ratio, the spread to standardize is the residual $\hat{u}_t = A_t - \hat{\alpha} - \hat{\beta}B_t$, not the $\beta = 1$ difference $A_t - B_t$. For the data in Table 7.4 the residuals are $0.1, -0.2, 0.5, -0.5, 0.2, -0.1$, with mean zero (as least-squares residuals always have) and sample standard deviation 0.346, so the final residual has a z-score of $z_6 = (-0.1-0)/0.346 \approx -0.29$. (Standardizing the raw difference instead would give $(3 - 2.5)/0.548 \approx 0.91$, even the opposite sign, which shows how much the choice of hedge ratio matters.) A common rule of thumb opens a trade when $|z_t|$ exceeds roughly 2 and closes it once z_t reverts near zero, so this spread, less than a third of a standard deviation from equilibrium, would not trigger a trade.

When the spread is unusually wide relative to its historical mean and standard deviation, a trader shorts the relatively expensive stock (here, A) and buys $\hat{\beta}$ shares of the relatively cheap one (B) for every share of A, betting that the spread will revert toward its mean; when the spread is unusually narrow, the trader takes the opposite position. Because the strategy trades the spread rather than either stock's direction, it is designed to profit whether the market rises or falls, a property called market neutrality, provided the cointegrating relationship remains stable. That assumption can silently break down, a risk Chapter 13 explores for statistical arbitrage strategies.

Pairs selection is also where the multiple-testing problem flagged in Section 7.3.1 bites hardest. A trader rarely tests one pair; a more typical screen runs the Engle-Granger test on every pair in a universe of, say, 20 bank stocks, which is $\binom{20}{2} = 190$ pairs. If no pair were cointegrated, a 5% test would still flag about $190 \times 0.05 = 9.5$ of them by chance, and those false positives are the pairs whose spreads stop mean-reverting once money is on them. Two standard defenses are the Bonferroni correction, which requires each individual test to clear $0.05/190 \approx 0.00026$ so that the chance of even one false positive stays below 5%, and the less conservative false discovery rate procedure of Benjamini and Hochberg, which instead controls the expected fraction of flagged pairs that are false. Neither replaces an economic reason to expect the relationship (same industry, same funding model, same regulator) and a test on data the screen never saw.

7.4.2 Vector Autoregression: Modeling Several Series Jointly

Every model so far in this chapter describes a single series in terms of its own past. A vector autoregression (VAR) extends the AR(p) framework to several series simultaneously, allowing each variable to depend on its own past values and the past values of every other variable in the system. A VAR(1) model for two series X_t and Y_t is

$$X_t = c_1 + a_{11}X_{t-1} + a_{12}Y_{t-1} + \varepsilon_{1,t},$$
$$Y_t = c_2 + a_{21}X_{t-1} + a_{22}Y_{t-1} + \varepsilon_{2,t},$$

where the off-diagonal coefficients a_{12} and a_{21} capture cross-series dependence, for example how last month's interest rate might help predict this month's exchange rate, and vice versa. This models interconnected series such as interest rates, exchange rates, and inflation together, rather than with separate univariate ARIMA models that ignore their interdependence.

Conveniently, given the lag order, each equation of the VAR(1) system can be estimated separately by OLS, regressing X_t on X_{t-1} and Y_{t-1} and separately regressing Y_t on the same two variables. The number of lags (VAR(p) for $p > 1$) is chosen by AIC or BIC as for ARIMA orders, since a VAR with too many lags for the data overfits as readily as an over-parameterized univariate model.

VAR models are also the standard tool for testing Granger causality: variable X is said to "Granger-cause" variable Y if past values of X improve the prediction of Y beyond what past values of Y alone provide, formally tested by checking whether the coefficients on lagged X in the Y equation (here, a_{21}) are statistically significant.

Granger causality is a statement about predictive usefulness, not causality in the structural sense of Chapter 1's discussion of correlation versus causation; a variable can Granger-cause another simply because it reflects the same information sooner.

Once a VAR is fitted, an impulse response function traces how a one-time shock to one variable propagates through the system: a hypothetical one-unit shock is applied to $\varepsilon_{1,t}$ (holding $\varepsilon_{2,t} = 0$), and the model is iterated forward to see how $X_{t+1}, X_{t+2}, \ldots$ and $Y_{t+1}, Y_{t+2}, \ldots$ respond.

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This is a standard tool of macroeconomic policy analysis, tracing for example how a shock to short-term rates reaches exchange rates and eventually inflation, and the multivariate analogue of the AR(1) multi-step forecast.

A concrete example makes the mechanism tangible. Suppose a VAR(1) has already been fitted (via the equation-by-equation OLS procedure above) to changes in a short-term interest rate X_t and an exchange rate Y_t, with estimated coefficients $a_{11} = 0.6$, $a_{12} = 0.2$, $a_{21} = 0.1$, and $a_{22} = 0.7$ (intercepts omitted, since an impulse response tracks the deviation from baseline rather than the level). A one-unit shock to the interest rate at time 0, with the exchange rate unshocked ($X_0 = 1$, $Y_0 = 0$), propagates as

$$X_1 = 0.6(1) + 0.2(0) = 0.6, \qquad Y_1 = 0.1(1) + 0.7(0) = 0.1,$$

$$X_2 = 0.6(0.6) + 0.2(0.1) = 0.38, \qquad Y_2 = 0.1(0.6) + 0.7(0.1) = 0.13,$$

and continuing this recursion gives

$$X_3 = 0.254, \qquad Y_3 = 0.129, \qquad X_4 = 0.178, \qquad Y_4 = 0.116.$$

Two features stand out. First, the exchange rate, though not shocked, responds at once, rising from 0 to 0.1 through the cross-series coefficient $a_{21} = 0.1$, a spillover that separate univariate ARIMA models could never capture. Second, its response does not fade alongside the interest rate's: it rises further, from 0.1 to 0.13, before declining, because it keeps absorbing the still-elevated interest rate for a second period. This lagged peak is the signature of cross-series transmission rather than independent reversion.

Both series eventually decay toward zero, which the individual coefficients alone do not guarantee; it depends on the stability of the whole coefficient matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

jointly. The multivariate generalization of the univariate AR(1) stationarity condition $|\phi| < 1$ is that every eigenvalue of A must lie strictly inside the unit circle. For this system the eigenvalues are 0.5 and 0.8, both below 1 in absolute value, so the impulse response must decay to zero rather than explode, as the values $0.6, 0.38, 0.254, 0.178$ for X and $0.1, 0.13, 0.129, 0.116$ for Y suggested.

A VAR fitted to non-stationary series (interest rate levels rather than changes, say) can easily have an eigenvalue at or above 1, and then any impulse response or multi-step forecast built from it, however precisely computed, describes an explosive process rather than an economic dynamic. This is the multivariate version of the unit-root problem the Dickey-Fuller test detects in a single series, and it is why a VAR, like an ARIMA model, is normally fit to differenced or otherwise stationary series.

7.4.3 Worked Example 7.3: PCA and Forecasting the Term Structure of Interest Rates

A VAR handles two or three series comfortably, but Chapter 5's yield curve is a whole curve, one yield for every maturity a government issues. Modeling eight or more maturities as one high-dimensional VAR is possible but wasteful, because adjacent points on the curve are very highly correlated: the 2-year and 3-year yields almost always move together.

Chapter 6 introduced principal component analysis (PCA), which finds the direction w maximizing $\mathrm{Var}(Xw)$. Applied to a panel of yield-curve maturities, it turns this redundancy into a forecasting advantage, since a few extracted factors carry nearly all of the curve's variation.

This worked example uses real data: the U.S. Treasury's daily par yield curve rates from January 2, 1990 through December 29, 2023, restricted to the eight maturities with essentially complete history over that period (3-month, 6-month, 1-year, 2-year, 3-year, 5-year, 7-year, and 10-year), giving 8,503 daily observations. Applying PCA to this panel, mean-centered but not rescaled (so that maturities with larger yield swings are not shrunk to match calmer ones), gives the explained variance ratios in Table 7.5.

The first three components explain 99.96% of the variation across 34 years of daily yield-curve moves: an eight-dimensional forecasting problem collapses, with almost no loss of information, to a three-dimensional one. This is one of the most reproduced results in fixed income, first documented by Litterman and Scheinkman

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Component	PC1	PC2	PC3	PC4	PC5
Variance explained	95.95%	3.78%	0.22%	0.03%	0.01%

Table 7.5: Explained variance ratio by principal component, 1990–2023 Treasury yield curve panel

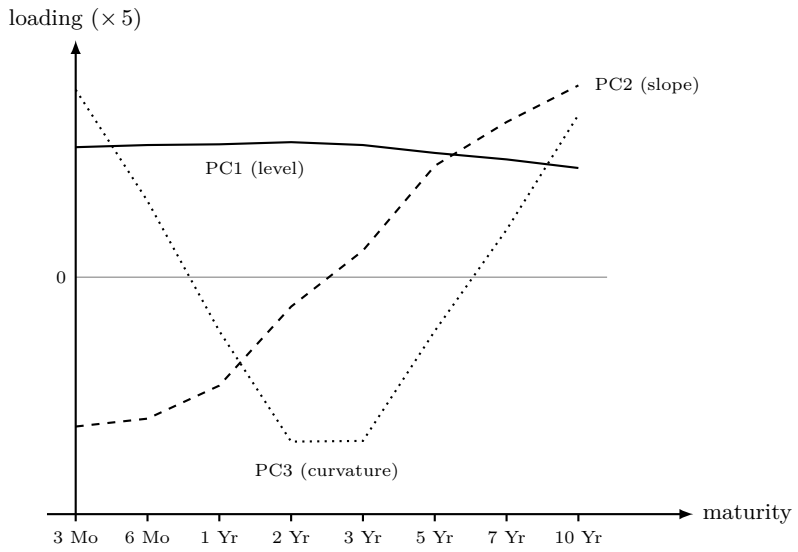


Figure 7.2: PCA loadings by maturity for the first three principal components of the 1990–2023 Treasury yield curve panel

(1991), and the components have well-known, interpretable shapes once their loadings, the weight of each maturity, are plotted against maturity.

Figure 7.2 shows this level-slope-curvature structure. PC1's loadings range only from 0.30 to 0.38 across all eight maturities, all positive and all close together: a positive PC1 shock raises every yield on the curve by roughly the same amount, a parallel shift, which is why PC1 is called the *level* factor. PC2's loadings move monotonically from -0.42 at the 3-month maturity to $+0.53$ at the 10-year maturity, so a positive PC2 shock lowers short-term yields while raising long-term ones, steepening the curve; this is the *slope* factor.

PC3's loadings are positive at both ends of the curve (0.52 at 3 months, 0.45 at 10 years) and negative in the middle (-0.46 at both 2 and 3 years), a hump-shaped pattern called the *curvature* factor, since a positive PC3 shock bends the belly of the curve down relative to both ends.

Level, slope, and curvature are themselves time series, obtained by projecting each day's curve onto the loading vectors, so they can be forecast with this chapter's univariate tools and mapped back to a curve-wide forecast through the loadings. Consider the dominant factor, PC1. Testing its level directly with the Augmented Dickey-Fuller test of Section 7.2.3 gives a test statistic of -2.295 ($p = 0.174$), failing to reject the unit-root null: like the stock prices of Section 7.4.1, the level factor is non-stationary and cannot be fed directly into an AR model. Differencing it once gives a test statistic of -16.474 ($p < 0.0001$), decisively stationary.

Fitting an AR(1) model to the differenced level factor ΔPC1_t by OLS gives

$$\Delta\text{PC1}_t = -0.0011 + 0.0411\,\Delta\text{PC1}_{t-1} + \varepsilon_t,$$

with the AR(1) coefficient statistically significant ($p = 0.0002$) thanks to the sample of 8,502 daily changes, but economically tiny: $R^2 = 0.0017$, so the previous day's move explains under one-fifth of one percent of the next. On the last trading day of the sample (December 29, 2023), the level factor stood at 3.063 after a daily change of -0.023; the fitted model forecasts the next change as $-0.0011 + 0.0411(-0.023) \approx -0.002$, an essentially negligible parallel shift. Mapped back through the PC1 loadings in Figure 7.2, this forecasts each

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of the eight maturities to move by only about -0.0007 to -0.0008 percentage points the next day, an order of magnitude smaller than a single basis point.

The result is informative, not disappointing. A significant but negligible R^2 is what an efficient, highly liquid market should produce: if daily changes in the Treasury curve's level were meaningfully predictable from their own history, large participants would trade the predictability away, the random-walk logic of Section 7.2.2 confirmed on 34 years of term-structure data. The payoff of PCA is not that it makes the level factor easy to predict; nothing makes daily interest-rate changes easy to predict. It replaces an unwieldy eight-maturity problem with a three-factor one, and whatever structure a more sophisticated model finds (a GARCH specification of Section 7.5.1 on ΔPC1_t's volatility, for instance) maps back to every maturity at once through the fixed loadings, instead of requiring eight separate models to stay mutually consistent.

7.5 Volatility and State-Space Models

Volatility is itself predictable in a way the level is not, and some of what matters is never observed directly.

7.5.1 Volatility Clustering and the GARCH Family

A defining stylized fact of financial returns is that their variance is not constant: large price moves tend to be followed by more large moves (of either sign), and calm periods tend to persist, a pattern called volatility clustering. This matters for the option pricing models of Chapter 5, which assumed a constant volatility σ; GARCH models let volatility evolve with the data instead.

The ARCH(1) model (autoregressive conditional heteroskedasticity) makes today's conditional variance a function of yesterday's squared shock,

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2, \qquad \omega > 0,\ \alpha \geq 0.$$

The GARCH(1,1) model, by far the most widely used volatility model in practice, adds a dependence on yesterday's own variance forecast,

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0,\ \alpha + \beta < 1. \tag{7.7}$$

The condition $\alpha + \beta < 1$ ensures the process is stationary, with a well-defined long-run variance $\omega/(1-\alpha-\beta)$ that the forecast reverts to over time; $\alpha + \beta$ close to 1, a common finding in financial data, means volatility shocks die out slowly, which is what volatility clustering looks like.

To make the recursion concrete, reuse the four AssetA returns from Chapter 2,

$$r = (0.0200, -0.0098, 0.0297, -0.0096),$$

whose sample variance was 0.000414. Take this as the seed σ_1^2, and suppose (for illustration, not estimated from only four data points) that $\omega = 0.00002$, $\alpha = 0.10$, and $\beta = 0.85$, so that $\alpha + \beta = 0.95$ and the long-run variance is $\omega/(1-0.95) = 0.0004$, a long-run volatility of 2.00%. The recursion needs shocks rather than returns: a GARCH model sits on a mean equation $r_t = \mu + \varepsilon_t$, so the shock is the return's deviation from its expected value, $\varepsilon_t = r_t - \mu$. Setting $\mu = 0$ lets the returns stand in for the shocks, the conventional treatment for daily data, where the average drift is two orders of magnitude smaller than a typical daily move and barely affects the fitted variance path. Applying the recursion with $\varepsilon_{t-1} = r_{t-1}$ on that basis gives Table 7.6.

The forecast volatility rises from 1.95% to 2.08% exactly after the large return of 2.97% at $t = 3$, and falls back toward the long-run 2.00% once smaller returns follow, the clustering GARCH is designed to capture. In practice ω, α, and β are estimated from a much longer history by maximum likelihood, but the recursion works identically once they are fitted.

The GARCH(1,1) model treats a positive and a negative return of the same magnitude as equally informative about future volatility, since the recursion depends only on r_{t-1}^2. This is a well-documented shortcoming: in equity markets especially, a large negative return is typically followed by higher volatility than an equally large positive one, an effect sometimes called the leverage effect.

Extensions such as EGARCH (exponential GARCH) and GJR-GARCH let the recursion respond to the sign of the previous shock as well as its size, at the cost of extra parameters. They are not developed here, but the symmetry built into GARCH(1,1) matters when reading its forecasts around downturns, where it is most likely to fail.

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t	Shock used, r_{t-1}	r_{t-1}^2	σ_t^2	σ_t (volatility)
1 (seed)	–	–	0.000414	2.03%
2	0.0200	0.00040	0.000412	2.03%
3	−0.0098	0.00010	0.000380	1.95%
4	0.0297	0.00088	0.000431	2.08%
5	−0.0096	0.00009	0.000396	1.99%

Table 7.6: GARCH(1,1) variance recursion using Chapter 2's AssetA returns

7.5.2 State-Space Models and the Kalman Filter

Suppose a fund re-estimates a stock's market beta from a rolling regression each quarter and finds the estimate bouncing around noticeably from one quarter to the next. Some of the bounce is estimation noise that would appear even if the true beta never changed, but some may be the true beta drifting as the business evolves, and a single quarter's estimate cannot tell the two apart.

The state-space representation models this kind of time-varying behavior by separating an unobserved ("latent") state that evolves over time from the observed data that carry noisy information about it:

$$\begin{aligned} \text{State equation:} \quad & \beta_t = \beta_{t-1} + \eta_t, \\ \text{Observation equation:} \quad & y_t = x_t^\top \beta_t + \varepsilon_t, \end{aligned}$$

where β_t is an unobserved parameter that is allowed to drift slowly over time (for example, a stock's market beta, or the level of an interest rate), and y_t is what is actually observed. The Kalman filter is the standard algorithm for estimating β_t recursively as observations arrive, updating a running estimate and its uncertainty at every step rather than re-estimating from scratch.

Each step has two parts: a prediction from the state equation alone, then an update that corrects it with the new observation, weighted by the relative reliability of prediction and observation. Concretely, given a current estimate $\hat{\beta}_{t-1}$ with variance P_{t-1}, a process (state) noise variance Q, and an observation noise variance R,

$$\begin{aligned} \text{Predict:} \quad & P_{t|t-1} = P_{t-1} + Q, \\ \text{Kalman gain:} \quad & K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R}, \\ \text{Update:} \quad & \hat{\beta}_t = \hat{\beta}_{t-1} + K_t(y_t - \hat{\beta}_{t-1}), \qquad P_t = (1 - K_t)P_{t|t-1}. \end{aligned}$$

Suppose a fund's current estimate of a stock's market beta is $\hat{\beta}_0 = 1.0$ with variance $P_0 = 0.04$, the true beta is believed to drift slowly with process variance $Q = 0.001$ per period, and a single period's regression-based beta estimate is noisy with observation variance $R = 0.09$. If the next two periods' noisy beta estimates are $y_1 = 1.30$ and $y_2 = 1.25$, the filter updates as follows: at $t = 1$, $P_{1|0} = 0.041$, $K_1 = 0.041/0.131 \approx 0.313$, giving $\hat{\beta}_1 = 1.0 + 0.313(1.30 - 1.0) \approx 1.094$ and $P_1 \approx 0.028$; at $t = 2$, $K_2 \approx 0.245$, giving $\hat{\beta}_2 = 1.094 + 0.245(1.25 - 1.094) \approx 1.132$. The estimate moves toward each noisy observation without jumping all the way to it, as exponential smoothing does with a fixed weight α; the Kalman filter is an adaptive generalization of exponential smoothing whose weight K_t adjusts to the uncertainty in the current estimate relative to the noise in new observations. The gain shrinks across the two updates, from 0.313 to 0.245, as confidence in the running estimate grows, while the process noise Q reintroduces some uncertainty each step so that the filter can still adapt to real changes in beta.

Deriving the update equations is beyond the scope of this chapter, but the idea that parameters can be tracked as they evolve is directly useful in finance: a fund's exposure to a risk factor, or an asset's volatility (an alternative to GARCH known as stochastic volatility modeling), can both be framed this way.

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7.6 Validation and Accuracy

Time order makes the standard validation split invalid, so this part replaces it and asks what a forecast should be scored against.

7.6.1 Walk-Forward Validation for Time Series

Chapter 6 noted that k-fold cross-validation can mislead for time series, because a random split can train a model on data from after the validation period, information never available when a real forecast is made. Walk-forward validation keeps the training window entirely before the validation window and rolls both forward, as in Figure 7.3.

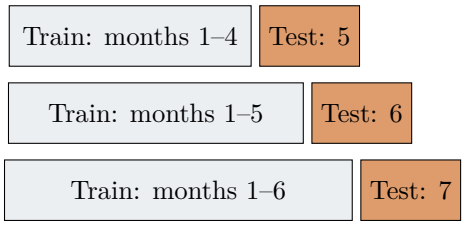


Figure 7.3: Walk-forward validation across three successive train/test splits

At each step, the model is refit (or updated) using only data available up to that point, produces a forecast for the next period, and that forecast is scored against the realized outcome once it arrives. The process rolls forward through the sample, and the reported performance is the average over these out-of-sample forecasts. It costs more than a single train-test split, since the model may be refit many times, but it is the only procedure that mimics how a forecasting model is used, one period at a time, without peeking ahead.

7.6.2 Evaluating Forecast Accuracy

The out-of-sample forecasts from walk-forward validation must be scored with a metric suited to the problem. Beyond the mean squared error used elsewhere in this book, two metrics are especially common in time series work. The root mean squared error (RMSE),

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(\hat{X}_t - X_t)^2},$$

returns the error to the original units of the series (unlike MSE, which is in squared units), making it directly interpretable, for example, as an average error of "2.1 basis points" rather than "4.4 squared basis points." The mean absolute percentage error (MAPE),

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{X_t - \hat{X}_t}{X_t}\right|,$$

expresses errors as a percentage of the actual value, which makes accuracy comparable across series of very different scales (dollar errors on a \$10 stock and a \$400 stock are not comparable, but percentage errors are). Its weakness is that it is undefined when $X_t = 0$ and unstable when X_t is near zero, a real concern for interest rates or spreads, which can pass through zero, unlike prices.

Beyond a single summary statistic, it is standard practice to compare a candidate model's walk-forward RMSE or MAPE against at least one naive benchmark, such as a forecast that simply repeats the last observed value ($\hat{X}_{t+1} = X_t$) or the simple moving average from Section 7.3.5. A sophisticated model that cannot beat such a benchmark walk-forward adds no predictive value, however compelling its in-sample fit.

A small worked comparison shows why this benchmark matters. Suppose a walk-forward evaluation produces five one-step-ahead forecasts for a price series with actual values $100, 102, 98, 105, 101$. A fitted model

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forecasts $99, 103, 97, 104, 102$, while the naive "repeat the last value" benchmark forecasts $98, 100, 102, 98, 105$ (each simply the prior period's actual value). Computing both metrics for each,

$$\text{RMSE}_{\text{model}} = 1.00, \qquad \text{RMSE}_{\text{naive}} = 4.22, \qquad \text{MAPE}_{\text{model}} = 0.99\%, \qquad \text{MAPE}_{\text{naive}} = 3.73\%,$$

the fitted model beats the naive benchmark by a wide margin, roughly a fourfold reduction in both metrics. That is encouraging, but five forecasts are far too few to show the model captures exploitable structure rather than luck. Whether one forecast's loss is significantly smaller than another's is what the Diebold-Mariano test formally assesses: it treats the period-by-period difference between the two models' losses as a series and tests whether its mean is zero, and with five such differences it has very little power. A long walk-forward history is what turns a gap like this into evidence. Had the two RMSE values been close, say 4.22 versus 4.05, the conclusion would be the opposite: the model's complexity buys almost nothing over repeating the last value, and a practitioner should ask whether it is worth the estimation risk and maintenance cost.

7.7 Challenges in Time Series Forecasting for Finance

Several challenges recur across almost every time series application in finance.

Regime changes and structural breaks. A model estimated over a calm period, including the AR(1) and GARCH parameters estimated above, can become badly miscalibrated once the underlying regime shifts, for example when a central bank changes policy or a market enters a crisis. A structural break is a sudden shift in the data-generating process itself, unlike the gradual change a stationary model allows. Formal tests detect breaks at a known or unknown date (the Chow and CUSUM tests are classical examples), but periodic re-estimation with walk-forward validation is usually a more robust defense than detecting every break in real time, since a break large enough to matter economically usually shows up as a sudden deterioration in walk-forward accuracy.

Limited effective sample size. Financial time series often look large (years of daily data) but contain far less independent information than the raw observation count suggests, because nearby observations are highly correlated. This limits how many parameters can be reliably estimated and is part of why simple, low-order models (AR(1), GARCH(1,1)) so often perform competitively against more elaborate alternatives.

Model selection versus overfitting. Adding autoregressive or moving-average terms always improves the in-sample fit. Information criteria such as AIC and BIC, with walk-forward validation, guard against a model that fits historical noise rather than a persistent pattern.

Fat tails and non-normality. Financial returns have more extreme observations than a normal distribution predicts. GARCH models are therefore often paired with a fatter-tailed error distribution (such as the Student's t), since a correctly time-varying variance does not by itself guarantee well-calibrated tail forecasts.

Multi-step forecasting compounds uncertainty. A one-step-ahead forecast, like this chapter's AR(1) example, is usually far more reliable than the same model's five- or twenty-step forecast, since uncertainty compounds with the horizon, and decisions resting on longer-horizon forecasts should treat them as considerably less certain.

Spurious cointegration and relationship breakdown. The pairs-trading strategy in Section 7.4.1 depends on a cointegrating relationship remaining stable, but with enough series to search, some pairs will appear cointegrated by chance, the multiple-testing problem again. Even an economically sound relationship can break down, for example if a merger, a change of strategy, or a regulatory shift changes one company's fundamentals relative to its peer.

Model misspecification in multivariate systems. A VAR model's conclusions, including any Granger-causality test, are sensitive to which variables are included and how many lags are used; omitting a variable that drives both series can produce a spurious Granger-causality finding, suggesting that one included variable predicts another when both are driven by the omitted one.

Classical models versus neural sequence models. Every model in this chapter, AR(1), ARIMA, GARCH, and VAR, is a classical statistical model with a small number of interpretable parameters, estimated by OLS or maximum likelihood. A recurrent neural network can in principle learn a far more flexible mapping from a series' history to its next value than any fixed linear, fixed-lag model, and Chapter 17 works through one, a hand-computed RNN forward pass. The flexibility is not free: such a model has many more parameters

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to fit from data that contain far less independent information than their observation count suggests, so it often costs more in overfitting than it returns in accuracy unless disciplined by walk-forward validation. The choice between a classical and a neural sequence model is therefore settled less by sophistication than by what a limited, noisy dataset can support without memorizing its own history.

The newest entrant in this debate sidesteps the limited-sample objection. **Time-series foundation models**, large neural sequence models pretrained once on enormous corpora of diverse time series (retail demand, energy load, traffic, weather, and much else) and then applied to a new series with little or no additional fitting, arrived in the mid-2020s under names like Chronos (Ansari et al. 2024) and TimesFM (Das et al. 2024), borrowing directly from the pretrain-then-adapt recipe Chapter 16 describes for language models. Because the parameter fitting happened on the pretraining corpus rather than the forecaster's short series, the overfitting arithmetic changes: the model brings pattern knowledge in, and on generic benchmarks it produces respectable zero-shot forecasts of unseen series. Nothing specific to finance has changed: published evaluations so far show these models competitive with, not dominant over, well-tuned classical baselines on financial series, whose low signal-to-noise ratio and regime shifts a mostly non-financial corpus does little to prepare for. Such a model is one more candidate to be judged by the same walk-forward, benchmark-against-naive discipline, not an exemption from it.

The companion notebook's optional final section applies that discipline to one such model, Chronos-Bolt-Small, a 48-million-parameter version of Chronos small enough to run on a laptop CPU, on the monthly 10-year yield of Section 7.3.7. For each month from January 2014 to December 2023 it forecasts the next month's yield zero-shot from all earlier months, the same walk-forward protocol used for the two benchmarks. Over the 120 forecasts its root mean squared error is 25.87 basis points, against 23.62 for a random walk and 23.64 for an AR(1) refit by OLS each month, and a Diebold-Mariano test rejects equal accuracy against both benchmarks ($p = 0.010$ and 0.014). Its 10%–90% forecast band covered 78.3% of the outcomes, close to the nominal 80%, so its uncertainty statements were sensible even though its point forecasts lost. One series and one decade settle nothing in general, but the outcome is the one the efficient-market reasoning of Section 7.4.3 predicts for a liquid rate: pattern knowledge imported from retail demand and electricity load does not beat a random walk on the 10-year Treasury yield.

7.8 A Practical Workflow for Building a Time Series Model

Figure 7.4 ties the chapter's tools together in the order they are typically applied.

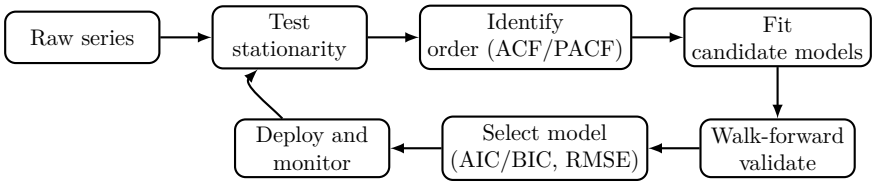


Figure 7.4: A practical time series modeling workflow

The first step is always to test stationarity (Section 7.2.3), since every model in this chapter, from AR(1) to GARCH to VAR, assumes stationarity in levels or after transformation; a non-stationary series is differenced (or converted to returns, as in Chapter 2) and re-tested.

Next, the ACF and PACF guide the choice of candidate orders, following the Box-Jenkins logic of the ARIMA section. Several candidates are fit and compared by information criteria (in-sample) and, more importantly, walk-forward validation (out-of-sample), scored by RMSE or MAPE and benchmarked against a naive forecast.

Only a model with walk-forward skill is deployed, and monitoring continues: the feedback arrow to the stationarity test reflects regime changes and structural breaks, which may require a model to be re-identified, not just re-fit, as data arrive.

The workflow applies to the single-series models (AR, ARIMA, GARCH) and the multi-series extensions (cointegration, VAR) alike; the multivariate case checks stationarity and lag orders for a whole system jointly.

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7.9 Key Takeaways

Time series methods extend the statistical toolkit of Chapter 6 to data where order matters. Stationarity makes a fitted model meaningful for the future, and checking for it, formally with the Dickey-Fuller test or informally with the correlogram, is the first step in any analysis. The AR(1) example showed coefficients estimated by the OLS logic of any regression, and the GARCH example showed volatility itself being modeled and forecast, feeding the option pricing and risk models of later chapters.

The cointegration and VAR material extended these ideas from a single series to relationships between series, previewing the pairs-trading strategies of Chapter 13 and the multivariate risk factor models of Chapter 8.

Walk-forward validation, scored by RMSE or MAPE and benchmarked against a naive forecast, follows from a principle repeated throughout this book: never evaluate a model with information it would not have had in real time, and never trust it merely for beating a strawman.

On the 8-observation running example, the Dickey-Fuller test and the ACF confidence band showed that formal significance is much harder to achieve in small samples than a chart or a plausible coefficient suggests, and that gap should make an analyst more cautious about small-sample conclusions. The same tests run on 408 months of the 10-year Treasury yield in Section 7.3.7 gave clear answers: a unit root in the level that could not be rejected, changes that were decisively stationary, and a partial autocorrelation function that pointed straight to an AR(1).

The AR(1), exponential smoothing, and Holt forecasts of the same series showed that reasonable modeling choices (mean reversion versus trend extrapolation) can give meaningfully different forecasts from identical data, and that choosing between them takes economic judgment, not just the best fit to the past.

Finally, the move from single series to relationships between series mirrors a pattern that recurs in this book: understanding one object in isolation is a prerequisite for, not a substitute for, understanding how it interacts with others, whether two cointegrated stocks here or the many assets of Chapter 12's portfolios.

7.10 Suggested Exercises

The exercises span the chapter's tools, from fitting and forecasting an AR(1) series to stationarity testing, comparing forecasting methods, and relationships between series. Several build on the worked examples, so your answers can be checked against the numbers in the text.

7.1 Explain, in your own words, why fitting an AR(1) model directly to a price series rather than a return series is likely to produce an unreliable model.

7.2 Using the data in Table 7.1, verify by hand that the sample mean is $\bar{X} = 6.25$ and that $\hat{\rho}_1 = 0.50$.

7.3 Add a ninth observation, $X_9 = 3.0$, to Table 7.1, refit the AR(1) model, and compare the new $\hat{\phi}$ and one-step-ahead forecast $\hat{X}_{10}$ to the values computed in the chapter.

7.4 Using the GARCH(1,1) recursion in Table 7.6, compute σ_6^2 if a fifth AssetA return of $r_5 = 0.035$ were observed.

7.5 Explain why $\alpha + \beta < 1$ is required for a GARCH(1,1) process to be stationary, and describe what would happen to the variance forecast if $\alpha + \beta = 1$ exactly.

7.6 Describe a concrete scenario in which k-fold cross-validation, as used in Chapter 6, would give a misleadingly optimistic evaluation of a time series forecasting model.

7.7 Discuss one reason a GARCH(1,1) model fit on five years of daily returns during a calm market might forecast volatility poorly at the onset of a financial crisis.

7.8 Using the Dickey-Fuller regression in Section 7.2.3, explain why $\hat{\gamma} = \hat{\phi} - 1$, and verify this relationship numerically using the AR(1) coefficient computed in Section 7.3.1.

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7.9 Explain why the Dickey-Fuller test failed to reject the unit-root null hypothesis in Section 7.2.3 even though the fitted AR(1) coefficient clearly indicated mean reversion, and describe what you would expect to happen to the test's conclusion if the same series had 200 observations instead of 8.

7.10 Using Table 7.4, compute the spread on a hypothetical Day 7 where Stock A trades at 56 and Stock B trades at 51, and explain, using the mean and standard deviation of the historical spread, whether a pairs trader would be more likely to open a new position or close an existing one on that day.

7.11 Explain, in your own words, the difference between Granger causality and true causal influence, and describe one financial scenario in which two series could Granger-cause each other without either one truly driving the other.

7.12 A retailer's monthly sales data shows a strong, repeating spike every December. Explain why an ordinary ARIMA model without a seasonal component would likely underperform a SARIMA model on this series, and describe what seasonal period s you would use.

7.13 Using the multi-step forecasting formulas in Section 7.3.3, compute the 4-step-ahead AR(1) forecast $\hat{X}_{12}$ from $X_8 = 4.0$, and verify it lies between the 3-step and 5-step forecasts already reported in Table 7.2.

7.14 Using Holt's linear trend method from Section 7.3.5 with $\alpha = 0.5$ and $\beta = 0.2$ (instead of $\alpha = 0.4$, $\beta = 0.3$), recompute the level and trend after processing $X_2 = 8.0$ through $X_4 = 7.0$ (three updates starting from $L_1 = 9.0$, $T_1 = -1.0$), and compare your intermediate values to those reported in the chapter.

7.15 Using the Kalman filter recursion in Section 7.5.2, compute a third update step assuming a new observation $y_3 = 1.20$ arrives, continuing from $\hat{\beta}_2 \approx 1.132$ and $P_2 \approx 0.022$ with the same $Q = 0.001$ and $R = 0.09$.

7.16 Explain, using the workflow in Figure 7.4, what step or steps you would need to repeat if a monitoring system detected that a deployed time series model's walk-forward forecast errors had grown substantially larger over the past quarter.

7.17 Using the confidence-band rule in Section 7.2.4 ($\pm 1.96/\sqrt{T}$), compute how many observations T would be needed for a sample autocorrelation of $\hat{\rho}_1 = 0.50$ to be considered statistically significant at the 95% level, and comment on how this compares to the 8 observations actually available in Table 7.1.

7.18 Using this chapter's VAR(1) impulse response example, compute X_1 and Y_1 following a one-unit shock to the exchange rate instead of the interest rate (that is, $X_0 = 0$, $Y_0 = 1$), and explain, in terms of the individual coefficients a_{12} and a_{21}, why the interest rate's one-period response to an exchange-rate shock turns out to be larger than the exchange rate's one-period response to an interest-rate shock was.

7.19 Using this chapter's RMSE/MAPE comparison, recompute both metrics for the naive benchmark if its forecasts were instead $99, 101, 99, 104, 100$ (closer to the actual values than the original naive forecasts), and discuss whether the fitted model would still show a clear advantage over this improved benchmark.

7.20 Using Section 7.3.4's Ljung-Box calculation, explain in your own words why a large Q statistic (and correspondingly small p-value) is evidence that a model's residuals still contain autocorrelation, and why this counts against the model even though the model's own R^2 or fitted coefficients might look reasonable on their own.

7.21 Section 7.3.4 found that AIC and BIC both favor the AR(2) model over the AR(1) model, yet also cautioned that this conclusion may simply reflect overfitting on only six observations. Describe a concrete piece of additional evidence (not just a lower in-sample AIC or BIC) that would make you more confident the AR(2) model is genuinely better specified, not just better fitting.

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7.22 Using the Treasury AR(1) fit of Section 7.3.7 ($\hat{c} = 0.048$, $\hat{\phi} = 0.986$), compute the implied long-run mean $\hat{c}/(1-\hat{\phi})$ and the 12-month-ahead forecast from the December 2023 yield of 3.88%, and state what fraction of the gap between the current yield and the long-run mean the forecast expects to close in a year. Explain why the Dickey-Fuller results in Table 7.3 should make you cautious about both numbers.

7.23 The pretrained forecaster in this chapter's challenges section had a higher RMSE than a random walk, and a Diebold-Mariano test rejected equal accuracy at $p = 0.010$. Explain what the test compares, why one-step-ahead forecasts allow the simple t-statistic on the loss differences without a correction for autocorrelation, and describe the evidence (series, horizons, periods) you would want before adopting such a model for a fixed-income desk.

7.24 **Challenge (real data).** Download at least five years of daily closing prices for a real, liquid stock (for example, via the `yfinance` library, as Chapter 2 and Chapter 5 already do) and compute its daily log returns. (a) Test the return series for a unit root using the augmented Dickey-Fuller test, as in Section 7.2.3, and contrast the statistical power of this test on your (likely thousands of) real observations with this chapter's own 8-observation toy example, where the test failed to reject a unit root despite a clearly mean-reverting fitted coefficient. (b) Fit a GARCH(1,1) model to the return series and plot the fitted conditional volatility alongside the raw returns, identifying at least one real historical event (an earnings surprise, a market-wide shock) that produced a visible volatility spike. (c) Compare your fitted $\alpha + \beta$ to the stationarity condition discussed in this chapter, and comment on whether real daily equity returns typically sit close to the $\alpha + \beta = 1$ boundary this chapter warns is a sign of near-nonstationary volatility.

7.11 Further Reading

- Tsay (2010), *Analysis of Financial Time Series*. Written specifically for financial applications, with extensive coverage of ARIMA, GARCH, and multivariate time series methods including cointegration and VAR models.
- Hamilton (1994), *Time Series Analysis*. The classic, mathematically rigorous graduate-level reference for the state-space and Kalman filtering material introduced only briefly in this chapter, alongside a thorough treatment of stationarity and unit-root testing.
- Engle (1982), "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." The original paper introducing ARCH models, for readers interested in the primary source behind the GARCH family developed in this chapter.
- Hyndman and Athanasopoulos (2021), *Forecasting*. A highly practical, freely available online text covering exponential smoothing, ARIMA, and forecast evaluation with an emphasis on applied forecasting workflows like the one in Figure 7.4.
- Engle and Granger (1987), "Co-integration and Error Correction: Representation, Estimation, and Testing." The original paper introducing the cointegration concept and the Engle-Granger two-step testing procedure used in Section 7.4.1.
- Sims (1980), "Macroeconomics and Reality." The influential paper introducing vector autoregression as an alternative to large-scale structural macroeconomic models, foundational to the VAR and Granger-causality material in this chapter.
- Ansari et al. (2024), "Chronos: Learning the Language of Time Series." The paper introducing one of the pretrained time-series foundation models discussed in this chapter's closing challenges, applying the language-model pretraining recipe to zero-shot forecasting.
- Litterman and Scheinkman (1991), "Common Factors Affecting Bond Returns." The original paper documenting the level, slope, and curvature factor structure of the yield curve reproduced in Section 7.4.3.

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